
Definition 0.1 (Function). A function or mapping $f : X \rightarrow Y$ is a relation between two sets X and Y such that for any $x \in X$, there exists exactly one $y \in Y$ such that $f(x) = y$.

$$\forall x \in X, \exists! y \in Y \text{ s.t. } y = f(x)$$

The set X is called the domain of f and the set Y is called the codomain of f .

A function can be more rigorously defined as $f \subset X \times Y$ where $\forall x \in X, \exists y \in Y$ s.t. $(x, y) \in f$ and $(x, y) \in f \wedge (x, z) \in f \implies y = z$ (I used Wikipedia to recall part of this definition).

Definition 0.2 (Onto). Let $f : X \rightarrow Y$ be a mapping. We say that f is onto or surjective if for every $y \in Y$, there exists some $x \in X$ such that $y = f(x)$.

Definition 0.3 (One-to-one). Let $f : X \rightarrow Y$ be a mapping. We say that f is one-to-one or injective if for $x_1, x_2 \in X$,

$$f(x_1) = f(x_2) \iff x_1 = x_2$$

Equivalently,

$$x_1 \neq x_2 \iff f(x_1) \neq f(x_2)$$

Remark. Functions that are one-to-one and onto are also called bijections.

Examples:

$f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x$ is both one-to-one and onto.

$g : \mathbb{R} \rightarrow \mathbb{R}^+ \cup \{0\}$ given by $g(x) = x^2$ is onto due to its definition, but is not injective.

$h : \mathbb{R} \rightarrow \mathbb{R}$ given by $h(x) = e^x$ is one-to-one, but is not onto $\mathbb{R}$.

Apply the Gram-Schmidt Orthonormalization Process to the following bases for $\mathbb{R}^n$, using vectors in the order they are given.

Exercise.

$$B = \{(3, 4), (1, 0)\}$$

Solution.

$$\mathbf{v}_1 := (3, 4)$$

$$\mathbf{v}_2 := (1, 0) - \frac{\langle (1, 0), (3, 4) \rangle}{\langle (3, 4), (3, 4) \rangle} (3, 4) = (1, 0) - \frac{3}{25} (3, 4) = (1, 0) - \left(\frac{9}{25}, \frac{12}{25} \right) = \left(\frac{16}{25}, -\frac{12}{25} \right)$$

$$\mathbf{w}_1 := \frac{(3, 4)}{\sqrt{3^2 + 4^2}} = \left(\frac{3}{5}, \frac{4}{5} \right)$$

$$\mathbf{w}_2 := \frac{\left(\frac{16}{25}, -\frac{12}{25} \right)}{\frac{\sqrt{16^2 + 12^2}}{25}} = \frac{\left(\frac{16}{25}, -\frac{12}{25} \right)}{\frac{4}{5}} = \frac{5}{4} \left(\frac{16}{25}, -\frac{12}{25} \right) = \left(\frac{4}{5}, -\frac{3}{5} \right)$$

The set $\{\mathbf{w}_1, \mathbf{w}_2\}$ forms an orthonormal basis for $\mathbb{R}^2$. □

Exercise.

$$B = \{(-1, 2), (1, 0)\}$$

Solution.

$$\mathbf{v}_1 := (-1, 2)$$

$$\mathbf{v}_2 := (1, 0) - \frac{\langle (1, 0), (-1, 2) \rangle}{\langle (-1, 2), (-1, 2) \rangle} (-1, 2) = (1, 0) - \frac{-1}{5} (-1, 2) = (1, 0) - \left(\frac{1}{5}, -\frac{2}{5} \right) = \left(\frac{4}{5}, \frac{2}{5} \right)$$

$$\mathbf{w}_1 := \frac{(-1, 2)}{\sqrt{5}} = \left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right)$$

$$\mathbf{w}_2 := \frac{\left(\frac{4}{5}, \frac{2}{5} \right)}{\frac{\sqrt{16+4}}{5}} = \frac{5}{\sqrt{20}} \left(\frac{4}{5}, \frac{2}{5} \right) = \left(\frac{4}{\sqrt{20}}, \frac{2}{\sqrt{20}} \right)$$

The set $\{\mathbf{w}_1, \mathbf{w}_2\}$ forms an orthonormal basis for $\mathbb{R}^2$. □

Exercise.

$$B = \{(0, 1), (2, 5)\}$$

Solution.

$$\mathbf{v}_1 := (0, 1)$$

$$\mathbf{v}_2 := (2, 5) - \frac{\langle (2, 5), (0, 1) \rangle}{\langle (0, 1), (0, 1) \rangle} (0, 1) = (2, 5) - \frac{5}{1} (0, 1) = (2, 5) - (0, 5) = (2, 0)$$

$$\mathbf{w}_1 = (0, 1)$$

$$\mathbf{w}_2 = (1, 0)$$

The set $\{\mathbf{w}_1, \mathbf{w}_2\}$ forms an orthonormal basis for $\mathbb{R}^2$. □

Exercise.

$$B = \{(4, -3), (3, 2)\}$$

Solution.

$$\mathbf{v}_1 := (4, -3)$$

$$\mathbf{v}_2 := (3, 2) - \frac{\langle (3, 2), (4, -3) \rangle}{\langle (4, -3), (4, -3) \rangle} (4, -3) = (3, 2) - \frac{6}{25} (4, -3) = (3, 2) - \left(\frac{24}{25}, -\frac{18}{25} \right) = \left(\frac{51}{25}, \frac{68}{25} \right)$$

$$\mathbf{w}_1 := \frac{(4, -3)}{\sqrt{16+9}} = \left(\frac{4}{5}, -\frac{3}{5} \right)$$

$$\mathbf{w}_2 := \frac{\left(\frac{51}{25}, \frac{68}{25} \right)}{\frac{\sqrt{51^2+68^2}}{25}} = \frac{\left(\frac{51}{25}, \frac{68}{25} \right)}{\frac{85}{25}} = \frac{25}{85} \left(\frac{51}{25}, \frac{68}{25} \right) = \left(\frac{51}{85}, \frac{68}{85} \right)$$

The set $\{\mathbf{w}_1, \mathbf{w}_2\}$ forms an orthonormal basis for $\mathbb{R}^2$. □

Exercise.

$$B = \{(2, 1, -2), (1, 2, 2), (2, -2, 1)\}$$

Solution.

$$\mathbf{v}_1 := (2, 1, -2)$$

$$\mathbf{v}_2 := (1, 2, 2) - \frac{\langle (1, 2, 2), (2, 1, -2) \rangle}{\langle (2, 1, -2), (2, 1, -2) \rangle} (2, 1, -2) = (1, 2, 2) - 0 = (1, 2, 2)$$

$$\mathbf{v}_3 := (2, -2, 1) - \frac{\langle (2, -2, 1), (2, 1, -2) \rangle}{\langle (2, 1, -2), (2, 1, -2) \rangle} (2, 1, -2) - \frac{\langle (2, -2, 1), (1, 2, 2) \rangle}{\langle (1, 2, 2), (1, 2, 2) \rangle} (1, 2, 2) = (2, -2, 1)$$

$$\mathbf{w}_1 := \frac{(2, 1, -2)}{\sqrt{4+1+4}} = \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3} \right)$$

$$\mathbf{w}_2 := \frac{(1, 2, 2)}{\sqrt{1+4+4}} = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right)$$

$$\mathbf{w}_3 := \frac{(2, -2, 1)}{\sqrt{4+4+1}} = \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right)$$

The set $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ forms an orthonormal basis for $\mathbb{R}^3$. □

Exercise.

$$B = \{(1, 0, 0), (1, 1, 1), (1, 1, -1)\}$$

Solution.

$$\mathbf{v}_1 := (1, 0, 0)$$

$$\mathbf{v}_2 := (1, 1, 1) - \frac{\langle (1, 1, 1), (1, 0, 0) \rangle}{\langle (1, 0, 0), (1, 0, 0) \rangle} (1, 0, 0) = (1, 1, 1) - (1, 0, 0) = (0, 1, 1)$$

$$\mathbf{v}_3 := (1, 1, -1) - \frac{\langle (1, 1, -1), (1, 0, 0) \rangle}{\langle (1, 0, 0), (1, 0, 0) \rangle} (1, 0, 0) - \frac{\langle (1, 1, -1), (0, 1, 1) \rangle}{\langle (0, 1, 1), (0, 1, 1) \rangle} (0, 1, 1) = (1, 1, -1) - (1, 0, 0) - (0, 1, 1) = (0, 0, -2)$$

$$\mathbf{w}_1 := (1, 0, 0)$$

$$\mathbf{w}_2 := \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

$$\mathbf{w}_3 := \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$

The set $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ forms an orthonormal basis for $\mathbb{R}^3$. □

Exercise.

$$B = \{(4, -3, 0), (1, 2, 0), (0, 0, 4)\}$$

Solution.

$$\mathbf{v}_1 := (4, -3, 0)$$

$$\mathbf{v}_2 := (1, 2, 0) - \frac{\langle (4, -3, 0), (1, 2, 0) \rangle}{\langle (4, -3, 0), (4, -3, 0) \rangle} (4, -3, 0) = (1, 2, 0) - \frac{2}{25} (4, -3, 0) = \left(\frac{33}{25}, \frac{44}{25}, 0 \right)$$

$$\mathbf{v}_3 := (0, 0, 4) - \frac{\langle (0, 0, 4), (4, -3, 0) \rangle}{\langle (4, -3, 0), (4, -3, 0) \rangle} (4, -3, 0) - \frac{\langle (0, 0, 4), \left(\frac{33}{25}, \frac{44}{25}, 0 \right) \rangle}{\langle \left(\frac{33}{25}, \frac{44}{25}, 0 \right), \left(\frac{33}{25}, \frac{44}{25}, 0 \right) \rangle} \left(\frac{33}{25}, \frac{44}{25}, 0 \right) = (0, 0, 4)$$

$$\mathbf{w}_1 := \frac{(4, -3, 0)}{\sqrt{25}} = \left(\frac{4}{5}, -\frac{3}{5}, 0 \right)$$

$$\mathbf{w}_2 := \frac{\left(\frac{33}{25}, \frac{44}{25}, 0 \right)}{\frac{\sqrt{33^2 + 44^2}}{25}} = \frac{25}{55} \left(\frac{33}{25}, \frac{44}{25}, 0 \right) = \left(\frac{3}{5}, \frac{4}{5}, 0 \right)$$

$$\mathbf{w}_3 := (0, 0, 1)$$

The set $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ forms an orthonormal basis for $\mathbb{R}^3$. □

Exercise.

$$B = \{(0, 1, 2), (2, 0, 0), (1, 1, 1)\}$$

Solution. Obviously the first two vectors are orthogonal.

$$\mathbf{v}_1 := (0, 1, 2)$$

$$\mathbf{v}_2 := (2, 0, 0)$$

$$\begin{aligned}\mathbf{v}_3 &:= (1, 1, 1) - \frac{\langle (1, 1, 1), (0, 1, 2) \rangle}{\langle (0, 1, 2), (0, 1, 2) \rangle} (0, 1, 2) - \frac{\langle (1, 1, 1), (2, 0, 0) \rangle}{\langle (2, 0, 0), (2, 0, 0) \rangle} (2, 0, 0) \\ &= (1, 1, 1) - \frac{3}{5} (0, 1, 2) - \frac{1}{2} (1, 0, 0) = \left(0, \frac{2}{5}, -\frac{1}{5}\right)\end{aligned}$$

$$\mathbf{w}_1 := \left(0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$$

$$\mathbf{w}_2 := (1, 0, 0)$$

$$\mathbf{w}_3 := \frac{(0, \frac{2}{5}, -\frac{1}{5})}{\frac{\sqrt{5}}{5}} = \left(0, \frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right)$$

The set $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ forms an orthonormal basis for $\mathbb{R}^3$. □

Exercise.

$$B = \{(0, 1, 1), (1, 1, 0), (1, 0, 1)\}$$

Solution.

$$\mathbf{v}_1 := (0, 1, 1)$$

$$\mathbf{v}_2 := (1, 1, 0) - \frac{\langle (1, 1, 0), (0, 1, 1) \rangle}{\langle (0, 1, 1), (0, 1, 1) \rangle} (0, 1, 1) = (1, 1, 0) - \left(0, \frac{1}{2}, \frac{1}{2}\right) = \left(1, \frac{1}{2}, -\frac{1}{2}\right)$$

$$\begin{aligned}\mathbf{v}_3 &:= (1, 0, 1) - \frac{\langle (1, 0, 1), (0, 1, 1) \rangle}{\langle (0, 1, 1), (0, 1, 1) \rangle} (0, 1, 1) - \frac{\langle (1, 0, 1), (1, \frac{1}{2}, -\frac{1}{2}) \rangle}{\langle (1, \frac{1}{2}, -\frac{1}{2}), (1, \frac{1}{2}, -\frac{1}{2}) \rangle} \left(1, \frac{1}{2}, -\frac{1}{2}\right) \\ &= (1, 0, 1) - \left(0, \frac{1}{2}, \frac{1}{2}\right) - \left(\frac{1}{3}, \frac{1}{6}, -\frac{1}{6}\right) = \left(\frac{2}{3}, -\frac{4}{6}, \frac{4}{6}\right)\end{aligned}$$

$$\mathbf{w}_1 := \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

$$\mathbf{w}_2 := \frac{(1, \frac{1}{2}, -\frac{1}{2})}{\sqrt{1 + \frac{1}{4} + \frac{1}{4}}} = \frac{2}{\sqrt{6}} \left(1, \frac{1}{2}, -\frac{1}{2}\right) = \left(\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right)$$

$$\mathbf{w}_3 := \frac{(\frac{2}{3}, -\frac{2}{3}, \frac{2}{3})}{\frac{\sqrt{12}}{3}} = \frac{3}{\sqrt{12}} \left(\frac{2}{3}, -\frac{2}{3}, \frac{2}{3}\right) = \left(\frac{2}{2\sqrt{3}}, -\frac{2}{2\sqrt{3}}, 2\sqrt{3}\right) = \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

The set $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ forms an orthonormal basis for $\mathbb{R}^3$. □

Exercise.

$$B = \{(3, 4, 0, 0), (-1, 1, 0, 0), (2, 1, 0, -1), (0, 1, 1, 0)\}$$

Solution.

$$\mathbf{v}_1 := (3, 4, 0, 0)$$

$$\mathbf{v}_2 := (-1, 1, 0, 0) - \frac{1}{25}(3, 4, 0, 0) = (-1, 1, 0, 0) - \left(\frac{3}{25}, \frac{4}{25}, 0, 0\right) = \left(-\frac{28}{25}, \frac{21}{25}, 0, 0\right)$$

$$\mathbf{v}_3 := (2, 1, 0, -1) - \frac{10}{25}(3, 4, 0, 0) - \frac{5}{7}\left(-\frac{28}{25}, \frac{21}{25}, 0, 0\right) = \left(\frac{8}{5}, -\frac{6}{5}, 0, -1\right)$$

$$\mathbf{v}_4 := (0, 1, 1, 0) - \frac{4}{25}(3, 4, 0, 0) - \frac{3}{7}\left(-\frac{28}{25}, \frac{21}{25}, 0, 0\right) - \frac{-6}{25}\left(\frac{8}{5}, -\frac{6}{5}, 0, -1\right) = \left(\frac{108}{125}, \frac{44}{125}, 1, -\frac{6}{25}\right)$$

$$\mathbf{w}_1 := \frac{(3, 4, 0, 0)}{\sqrt{25}} = \left(\frac{3}{5}, \frac{4}{5}, 0, 0\right)$$

$$\mathbf{w}_2 := \frac{\left(-\frac{28}{25}, \frac{21}{25}, 0, 0\right)}{\frac{7}{5}} = \left(-\frac{4}{5}, \frac{3}{5}, 0, 0\right)$$

$$\mathbf{w}_3 := \frac{\left(\frac{8}{5}, -\frac{6}{5}, 0, -1\right)}{\sqrt{5}} = \left(\frac{8}{5\sqrt{5}}, -\frac{6}{5\sqrt{5}}, 0, -\frac{1}{\sqrt{5}}\right)$$

$$\mathbf{w}_4 := \left(\frac{108\sqrt{1205}}{6025}, \frac{44\sqrt{1205}}{6025}, \frac{5\sqrt{1205}}{241}, -\frac{6\sqrt{1205}}{1205}\right)$$

The set $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4\}$ hopefully forms an orthonormal basis for $\mathbb{R}^4$. □